JAMES MCCALLUM

Data Engineer | Staff Level

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PROFESSIONAL SUMMARY

With 10 years of hands-on experience building and maintaining large-scale data infrastructure, I've spent the last four years at staff level leading technical decisions for data platforms processing hundreds of terabytes daily. I'm passionate about designing systems that let analysts and scientists focus on insights rather than wrestling with data plumbing. My background spans both startup environments where I wore multiple hats and enterprise settings where I've managed complex pipelines serving hundreds of internal users.

WORK EXPERIENCE

Staff Data Engineer

Helix Analytics

helix-analytics.io

Glasgow, United Kingdom

March 2022 - Present

Led the redesign of our core ETL infrastructure, reducing query latencies by 65% and cutting cloud costs by £180k annually through optimized resource allocation and partition strategies. Mentored a team of four junior and mid-level engineers, conducting code reviews and designing training programs for new data technologies we adopted. Architected a real-time streaming pipeline using Kafka and Flink that processes 2 million events per second for our customer analytics platform, enabling live dashboards for enterprise clients. Established data quality frameworks and monitoring systems that reduced production incidents by 78% over 18 months.

Senior Data Engineer

Velocity Solutions Ltd

velocity-solutions.net

Edinburgh, United Kingdom

June 2018 - February 2022

Owned the migration of 15 legacy SQL Server databases to a cloud-native data warehouse on Snowflake, completed on schedule and within budget while maintaining zero downtime for critical business processes. Developed Python-based data orchestration tools using Airflow that automated 300+ manual data workflows, freeing up the analytics team to focus on higher-value work. Designed and implemented comprehensive logging and monitoring across all data pipelines using ELK stack, providing visibility that enabled faster incident response. Collaborated with data scientists to optimize machine learning model training pipelines, reducing training time from 6 hours to 24 minutes through caching strategies and distributed processing.

Data Engineer

Insight Data Systems

Glasgow, United Kingdom

August 2014 - May 2018

Built and maintained Hadoop clusters supporting the analytics team's ad-hoc queries and batch processing jobs, managing capacity planning and performance tuning across 50+ nodes. Developed SQL transformations and data models that standardized how raw business data was transformed into consumable analytics datasets.

Implemented automated testing frameworks for data pipeline outputs, catching data quality issues before they impacted downstream reporting. Partnered with the business intelligence team to understand requirements and deliver data infrastructure that supported their reporting needs.

EDUCATION

Master of Science in Computer Science

University of Edinburgh

Edinburgh, United Kingdom

2012 - 2014

Dissertation focused on distributed data processing algorithms

Bachelor of Science in Information Technology

Heriot-Watt University

Edinburgh, United Kingdom

2008 - 2012

TECHNICAL SKILLS

Data Warehousing and Processing: Snowflake, BigQuery, Apache Spark, Hadoop, Airflow, dbt

Programming Languages: Python, SQL, Scala, Bash

Streaming and Real-Time: Apache Kafka, Apache Flink, Kinesis

Cloud Platforms: AWS, Google Cloud Platform, Azure

Databases: PostgreSQL, MySQL, Redshift, Hive, Cassandra

Tools and Frameworks: Git, Jenkins, Docker, Kubernetes, Terraform

Monitoring and Logging: ELK Stack, Datadog, CloudWatch, Grafana

Version Control and CI/CD: GitHub, GitLab, Jenkins

CERTIFICATIONS

AWS Certified Solutions Architect - Professional (2021)

Certified Kubernetes Administrator (2020)